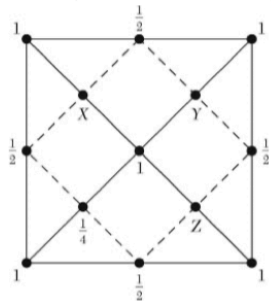


▶ PROBLEMS (* DENOTES DIFFICULT PROBLEMS)

FOR SECTION 1.2 BASIC CRYSTAL STRUCTURES

1. (a) What is the distance between nearest neighbors in silicon?
   (b) Find the number of atoms per square centimeter in silicon in the (100), (110), and (111) planes.
2. If we project the atoms in a diamond lattice onto the bottom surface with the heights of the atoms in unit of the lattice constant shown in the figure below; find the heights of the three atoms (X, Y, Z) on the figure.



3. Find the maximum fraction of the unit cell volume, which can be filled by identical hard spheres in the simple cubic, face-centered cubic, and diamond lattices.
4. Calculate the tetrahedral bond angle, the angle between any pair of the four bonds in a diamond lattice. (Hint: represent the four bonds as vectors of equal length. What must the sum of the four vectors equal? Take components of this vector equation along the direction of one of these vectors.)
5. If a plane has intercepts at $2(a)$, $3(a)$, and $4(a)$ along the three Cartesian coordinates, where a is the lattice constant, find the Miller indices of the planes.
6. (a) Calculate the density of GaAs (the lattice constant of GaAs is 5.65 Å, and the atomic weights of Ga and As are 69.72 and 74.92 g/mol, respectively).
   (b) A gallium arsenide sample is doped with tin. If the tin displaces gallium atoms in crystal lattice, are donors or acceptors formed? Why? Is the semiconductor n- or p-type?

FOR SECTION 1.4 ENERGY BANDS

7. The variation of silicon and GaAs bandgaps with temperature can be expressed as $E_g(T) = E_g(0) - \alpha T^2/(T + \beta)$, where $E_g(0) = 1.17$ eV, $\alpha = 4.73 \times 10^{-4}$ eV/K, and $\beta = 636$ K for silicon; and $E_g(0) = 1.519$ eV, $\alpha = 5.405 \times 10^{-4}$ eV/ K, and $\beta = 204$ K for GaAs. Find the bandgaps of Si and GaAs at 100 K and 600 K.

FOR SECTION 1.5 INTRINSIC CARRIER CONCENTRATION

8. Derive Eq. 14. (Hint: In the valence band, the probability of occupancy of a state by a hole is $[1 - F(E)]$.)
9. At room temperature (300 K) the effective density of states in the valence band is $2.66 \times 10^{19}\,\text{cm}^{-3}$ for silicon and $7 \times 10^{18}\,\text{cm}^{-3}$ for gallium arsenide. Find the corresponding effective masses of holes. Compare these masses with the free-electron mass.

10. Calculate the location of E_i in silicon at liquid nitrogen temperature (77 K), at room temperature (300 K), and at 100°C (let $m_p = 1.0\, m_0$ and $m_n = 0.19\, m_0$). Is it reasonable to assume that E_i is in the center of the forbidden gap?

11. Find the kinetic energy of electrons in the conduction band of a nondegenerate *n*-type semiconductor at 300 K.

12. (a) For a free electron with a velocity of 10^7 cm/s, what is its de Broglie wavelength? (b) In GaAs, the effective mass of electrons in the conduction band is $0.063\, m_0$. If they have the same velocity, find the corresponding de Broglie wavelength.

13. The intrinsic temperature of a semiconductor is the temperatures at which the intrinsic carrier concentration equals the impurity concentration. Find the intrinsic temperature for a silicon sample doped with 10^{15} phosphorus atoms/cm^3.

FOR SECTION 1.6 DONORS AND ACCEPTORS

14. A silicon sample at $T = 300$ K contains an acceptor impurity concentration of $N_A = 10^{16}$ cm^{-3}. Determine the concentration of donor impurity atoms that must be added so that the silicon is *n*-type and the Fermi energy is 0.20 eV below the conduction band edge.

15. Draw a simple flat energy band diagram for silicon doped with 10^{16} arsenic atoms/cm^3 at 77 K, 300 K, and 600 K. Show the Fermi level and use the intrinsic Fermi level as the energy reference.

16. Find the electron and hole concentrations and Fermi level in silicon at 300 K (a) for 1×10^{15} boron atoms/cm^3 and (b) for 3×10^{16} boron atoms/cm^3 and 2.9×10^{16} arsenic atoms/cm^3.

17. A Si sample is doped with 10^{17} As atoms/cm^3. What is the equilibrium hole concentration p_o at 300 K? Where is E_F relative to E_i?

18. A Ge sample is doped with 2.5×10^{13} cm^{-3} donor atoms that can be assumed to be fully ionized at room temperature. What is the free electron concentration for this sample at room temperature? (The intrinsic carrier concentration n_i of Ge is 2.5×10^{13} cm^{-3}.)

19. A *p*-type Si is doped with N_A acceptors close to the valence band edge. A certain type of donor impurity whose energy level is located at the intrinsic level is to be added to the semiconductor to obtain perfect compensation. If we assume that simple Fermi-level statistics apply, what is the concentration of donors required? Furthermore, after adding the donor impurity, what is the total number of ionized impurities if the above sample is perfect compensation?

20. Calculate the Fermi level of silicon doped with 10^{15}, 10^{17}, and 10^{19} phosphorus atoms/cm^3 at room temperature, assuming complete ionization. From the calculated Fermi level, check if the assumption of complete ionization is justified for each doping. Assume that the ionized donors is given by $n = N_D\left[1 - F(E_D)\right] = \dfrac{N_D}{1 + \exp[(E_F - E_D)/kT]}$

.

21. For an *n*-type silicon sample with 10^{16} cm^{-3} phosphorous donor impurities and a donor level at $E_D = 0.045$ eV, find the ratio of the neutral donor density to the ionized donor density at 77 K where the Fermi level is 0.0459 below the bottom of the conduction band. The expression for ionized donors is given in Prob. 20.

1. (a) What is the distance between nearest neighbors in silicon?

(b) Find the number of atoms per square centimeter in silicon in the (100), (110), and (111) planes.

Step 0｜必要背景

- 矽是**鑽石結構**：一個立方體邊長 a（晶格常數 5.43 Å），原子位置 = fcc 的 4 顆 + 錯開 $\left(\frac{a}{4}, \frac{a}{4}, \frac{a}{4}\right)$ 的另外 4 顆。
- 關鍵直覺：每顆原子都用 4 根鍵接到 4 個鄰居，位置像正四面體，所以鍵長一定和體對角線有關。

(a) 座標標 (單位a)

fcc 那組：$(0,0,0)\cdot(0,\frac{1}{2},\frac{1}{2})\cdot(\frac{1}{2},0,\frac{1}{2})\cdot(\frac{1}{2},\frac{1}{2},0)$

錯開那組：上面每個座標加 $(\frac{1}{4},\frac{1}{4},\frac{1}{4})$

最近的兩顆是 $(0,0,0)$ 和 $(\frac{1}{4},\frac{1}{4},\frac{1}{4})$

$$d = a\sqrt{(\tfrac{1}{4})^2+(\tfrac{1}{4})^2+(\tfrac{1}{4})^2} = \frac{\sqrt{3}}{4}a \approx 2.35\text{Å}$$

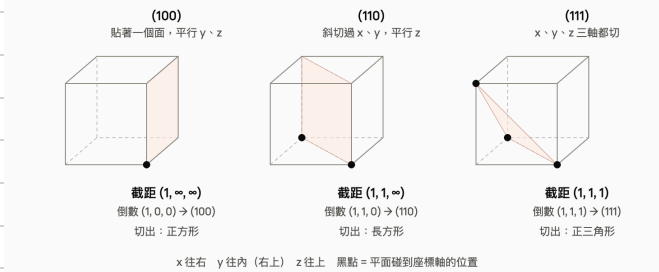
(b)

$$\text{面密度} = \frac{\text{一個重複單元內的等效原子數}}{\text{該重複單元的面積}}$$

先補基本功：Miller 指數在切什麼

(hkl) 表示一刀切過立方體，在 x, y, z 軸上的截距是 $\frac{a}{h}, \frac{a}{k}, \frac{a}{l}$。

- (100)：只切 x 軸，切出來是**正方形**（立方體的一個面）
- (110)：切 x、y 軸，平行 z 軸，切出來是**長方形**
- (111)：三軸都切，切出來是**正三角形**



，切法就一句話：**先決定平面碰到 x, y, z 三根軸的位置，再用碰到的點把平面連起來。**

※只算一次乘上係數

Step 1

$$\frac{1}{a^2} = \frac{1}{(5.43\times10^{-8}\ \text{cm})^2} = 3.392\times10^{14}\ \text{cm}^{-2}$$

Step 2

面	係數	大約多少
(100)	2	2
(110)	$2\sqrt{2}$	2.83
(111)	$4/\sqrt{3}$	2.31

Step 3

(100)：$2\times3.392\times10^{14} = 6.78\times10^{14}$

(110)：$2.83\times3.392\times10^{14} = 9.59\times10^{14}$

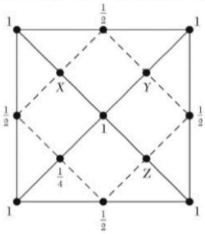
(111)：$2.31\times3.392\times10^{14} = 7.83\times10^{14}$

通式

$$\text{面密度} = \frac{N}{A} = \frac{N}{k\,a^2} = \underbrace{\frac{N}{k}}_{\text{係數}}\cdot\frac{1}{a^2}$$

其中 N 是等效原子數，面積 $A = k\,a^2$。

2. If we project the atoms in a diamond lattice onto the bottom surface with the heights of the atoms in unit of the lattice constant shown in the figure below; find the heights of the three atoms (X, Y, Z) on the figure.



fcc 那組原子高度只有 0, ½, 1；錯開的那組是 ¼, ¾。互相成鍵的原子，高度差是 ¼

Step 1

已知左下內部原子是 ¼。四顆內部原子規律「對角同高」，所以：

$X = \frac{3}{4}$, $Y = \frac{1}{4}$, $Z = \frac{3}{4}$

※ Check X 角邊中點 (½) 差 ¼；Y 角中心差 ¼；Z 角角落 ¼

6. (a) Calculate the density of GaAs (the lattice constant of GaAs is 5.65 Å, and the atomic weights of Ga and As are 69.72 and 74.92 g/mol, respectively).

(b) A gallium arsenide sample is doped with tin. If the tin displaces gallium atoms in crystal lattice, are donors or acceptors formed? Why? Is the semiconductor *n*- or *p*-type?

(a) 閃鋅礦結構一個單位晶胞有 4 顆 Ga + 4 顆 As：

$$\rho = \frac{4(M_{Ga} + M_{As})/N_A}{a^3} = \frac{4(69.72 + 74.92)/6.022\times10^{23}}{(5.65\times10^{-8})^3} = \frac{9.61\times10^{-22}}{1.80\times10^{-22}} \approx 5.33\ g/cm^3$$

(b)

Sn 是 IV 族，取代 III 族 Ga，多出 1 個電子：

形成 donor，半導體 n 型

FOR SECTION 1.5 INTRINSIC CARRIER CONCENTRATION

8. Derive Eq. 14. (Hint: In the valence band, the probability of occupancy of a state by a hole is $[1 - F(E)]$.)

<Sol> 電洞 = 沒被電子佔據，所以佔據機率是 1 - F(E)

$$p = \int_{-\infty}^{E_V} N(E)[1 - F(E)]dE, \quad N(E) = \frac{4\pi(2m_p)^{3/2}}{h^3}\sqrt{E_V - E}$$

非簡併 ($E_F - E \gg kT$) 時

$$1 - F(E) = \frac{1}{1 + e^{(E_F - E)/kT}} \approx e^{-(E_F - E)/kT}$$

令 $x = \frac{E_V - E}{kT}$，則 $E_F - E = (E_F - E_V) + xkT$

$$p = \frac{4\pi(2m_p)^{3/2}}{h^3}(kT)^{3/2} e^{-(E_F - E_V)/kT} \int_0^\infty \sqrt{x}\, e^{-x}dx \qquad \Gamma(3/2) = \sqrt{\pi}/2$$

$$p = 2\left(\frac{2\pi m_p kT}{h^2}\right)^{3/2} e^{-(E_F - E_V)/kT} = N_V e^{-(E_F - E_V)/kT}$$

9. At room temperature (300 K) the effective density of states in the valence band is $2.66 \times 10^{19}\,cm^{-3}$ for silicon and $7 \times 10^{18}\,cm^{-3}$ for gallium arsenide. Find the corresponding effective masses of holes. Compare these masses with the free-electron mass.

<Sol>

$$N_V = 2\left(\frac{2\pi m_p kT}{h^2}\right)^{3/2} \propto m_p^{3/2}$$

$$\frac{m_p}{m_0} = \left(\frac{N_V}{2.51\times10^{19}}\right)^{2/3}$$

$$Si: \left(\frac{2.66}{2.51}\right)^{2/3} \approx 1.04 \Rightarrow m_p \approx 1.04\, m_0$$

$$GaAs: \left(\frac{0.7}{2.51}\right)^{2/3} \approx 0.43 \Rightarrow m_p \approx 0.43 m_0$$

10. Calculate the location of E_i in silicon at liquid nitrogen temperature (77 K), at room temperature (300 K), and at 100°C (let $m_p = 1.0\, m_0$ and $m_n = 0.19\, m_0$). Is it reasonable to assume that E_i is in the center of the forbidden gap?

令 n=p 得到相对禁带中央的偏移

$$E_i = \frac{E_c + E_v}{2} + \frac{3}{4} kT \ln \frac{m_p}{m_n}$$

代入 $m_p/m_n = 1/0.19$. $\ln(1/0.19) = 1.661$. 代入 $\frac{3}{4}\times 1.661 = 1.246$.

Temperature	kT (eV)	$E_i - E_{mid} = 1.246\,kT$
77 k	0.00664	8.3 meV
300k	0.02585	32.2 meV
373k	0.03215	40.0 meV

11. Find the kinetic energy of electrons in the conduction band of a nondegenerate *n*-type semiconductor at 300 K.

动能 $= E - E_c$

$$\langle E - E_c \rangle = \frac{\int_0^\infty x\sqrt{x}\, e^{-x}\, dx}{\int_0^\infty \sqrt{x}\, e^{-x}\, dx} kT = \frac{\Gamma(5/2)}{\Gamma(3/2)} kT$$

$$KE = \frac{3}{2} kT = 1.5 \times 0.02585 \approx 0.0388\,eV \quad (300k)$$

FOR SECTION 1.6 DONORS AND ACCEPTORS

14. A silicon sample at $T = 300$ K contains an acceptor impurity concentration of $N_A = 10^{16}\,\text{cm}^{-3}$. Determine the concentration of donor impurity atoms that must be added so that the silicon is n-type and the Fermi energy is 0.20 eV below the conduction band edge.

$E_C - E_F = 0.20\,eV$ 决定 n，再由电中性倒推 N_D：

$$n = N_C e^{-(E_C - E_F)/kT} = 2.86\times10^{19} e^{-0.20/0.02585} \approx 1.25\times10^{16}\ \text{cm}^{-3}$$

n 型时 $p = n_i^2/n \approx 10^4 \ll n$，可忽略，电中性变成 $n + N_A = N_D$：

$$N_D = n + N_A = 1.25\times10^{16} + 1\times10^{16} \approx 2.25\times10^{16}\ \text{cm}^{-3}$$

16. Find the electron and hole concentrations and Fermi level in silicon at 300 K (a) for 1×10^{15} boron atoms/cm^3 and (b) for 3×10^{16} boron atoms/cm^3 and 2.9×10^{16} arsenic atoms/cm^3.

(a) $N_A = 10^{15} \gg n_i$，所以 $p \approx 10^{15}\,\text{cm}^{-3}$

$$n = \frac{n_i^2}{p} \approx 9.3\times10^{4}\,\text{cm}^{-3}.\quad E_i - E_F = kT\ln\frac{p}{n_i} \approx 0.299\,eV$$

E_F 在 E_i 下方约 0.30 eV（靠近价带，p 型）

(b) 净掺杂浓度 $N_A - N_D = 3\times10^{16} - 2.9\times10^{16} = 10^{15}$，所以结果与 (a) 相同

$p = 10^{15}$，$n \approx 9.3\times10^{4}$，$E_i - E_F \approx 0.30\,eV$

17. A Si sample is doped with 10^{17} As atoms/cm^3. What is the equilibrium hole concentration p_o at 300 K? Where is E_F relative to E_i?

$N_D = 10^{17} \gg n_i$，$\therefore n \approx 10^{17}$：

$$p_0 = \frac{n_i^2}{n} \approx 9.3\times10^{2}\,\text{cm}^{-3},\quad E_F - E_i = kT\ln\frac{n}{n_i} \approx 0.418\,eV$$

E_F 在 E_i 上方约 0.42 eV（靠近导带）

18. A Ge sample is doped with 2.5×10^{13} cm^{-3} donor atoms that can be assumed to be fully ionized at room temperature. What is the free electron concentration for this sample at room temperature? (The intrinsic carrier concentration n_i of Ge is 2.5×10^{13} cm^{-3}.)

由 $n = p + N_D$ 和 $np = n_i^2$ 得 $n^2 - N_D n - n_i^2 = 0$

$$n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i^2} = 1.25\times10^{13} + 2.80\times10^{13}$$

$$n \approx 4.05\times10^{13}\ \text{cm}^{-3}$$

檢查：$p = n_i^2/n \approx 1.54\times10^{13}$，且 $n - p = 2.5\times10^{13} = N_D$

20. Calculate the Fermi level of silicon doped with 10^{15}, 10^{17}, and 10^{19} phosphorus atoms/cm^3 at room temperature, assuming complete ionization. From the calculated Fermi level, check if the assumption of complete ionization is justified for each doping. Assume that the ionized donors is given by $n = N_D[1 - F(E_D)] = \dfrac{N_D}{1 + \exp[(E_F - E_D)/kT]}$

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假設完全游離使 $n = N_D$，求 $E_C - E_F = kT \ln(N_C/n)$，再驗証游離度比例.

用21題的 $E_C - E_D = 0.045$ eV

$$\frac{N_D^+}{N_D} = \frac{1}{1 + e^{(E_F - E_D)/kT}}$$

N_D (cm^{-3})	$E_C - E_F$	$E_F - E_D$	游離度比例	假設成立
10^{15}	0.265 eV	−0.220 eV	≈ 99.98 %	✓
10^{17}	0.146 eV	−0.101 eV	≈ 98 %	✓
10^{19}	0.027 eV	+0.018 eV	≈ 33 %	✗

21. For an n-type silicon sample with 10^{16} cm^{-3} phosphorous donor impurities and a donor level at $E_D = 0.045$ eV, find the ratio of the neutral donor density to the ionized donor density at 77 K where the Fermi level is 0.0459 below the bottom of the conduction band. The expression for ionized donors is given in Prob. 20.

游離度施体 $N_D^+ = \dfrac{N_D}{1 + e^{(E_F - E_D)/kT}}$，中性施体 $N_D^0 = N_D - N_D^+$，兩者相除：

$$\frac{N_D^0}{N_D^+} = e^{(E_F - E_D)/kT}$$

$E_F - E_D = 0.045 - 0.0459 = -0.0009$ eV，77 K 時 ≈ 0.00664 eV.

$$\frac{N_D^0}{N_D^+} = e^{-0.0009/0.00664} \approx 0.87$$